Validation of a finite-element solver for one-dimensional viscous sheet retraction

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Abstract

We validate a finite-element solver for the one-dimensional (long-wave) retraction of a viscous liquid sheet against an independent finite-difference implementation and against the asymptotic results of Ahsan *et al.* (arXiv:2507.04603). The two solvers share no code, no discretisation and no coordinate system, and agree to between 2×10^{-5} and 1.3×10^{-4} in the retraction speed across four decades of time. Measured against the published asymptotics, the early-time law and the late-time finite-sheet law are reproduced without adjustable constants, as are the tip-thickness relation and the crossover time $T_c = \mathcal{L}$. In the late-time infinite-sheet regime we measure a speed deficit $1 - \mathrm{d}X_c/\mathrm{d}T \to \frac{3}{2}(1+T)^{-2}$, against a published coefficient of unity. A third solver, written to share no coordinate, equation or boundary-condition type with the other two, returns the same value, and the limit is stable to 2×10^{-5} under variation of the fit window. The coefficient is shown to depend on the initial condition, so 3/2 is the value for the canonical rupture state rather than a universal constant.

1 Scope and nomenclature

Two solvers are compared throughout.

TC1D-FEM is the solver under validation: a finite-element discretisation of the long-wave equations in a Lagrangian mass coordinate, with the sheet thickness h on a cell-constant space and the velocity u on continuous piecewise-linear elements, integrated with BDF2. The free edge is a material point of the mesh, so no interface tracking or remeshing is required.

ABDS-CN is an independent reference: a transcription into Python of the finite-difference solver released with Ahsan *et al.*, which advances a Crank–Nicolson discretisation on a static grid with Picard iteration and a tridiagonal solve. Appendix A establishes that the transcription reproduces the original code to the precision at which the original can be read.

Two further solvers appear: TC1D-MASS, introduced in §4 to test one result independently, and TC1D-ALE, which solves the same model as TC1D-FEM but on a moving mesh in the physical coordinate. Table 1 sets out how the four differ.

The asymptotic predictions of Ahsan *et al.* are treated here as hypotheses to be tested, and are plotted as such: they are not used to normalise or calibrate any solver.

1.1 Units

Ahsan *et al.* scale on the half-thickness $h_0 = H/2$, with $u_{TC} = \sqrt{\gamma/\rho h_0}$, Ohnesorge number $Oh_p = \mu/\sqrt{\rho h_0 \gamma}$ and visco-inertial time $t_I = 4\mu h_0/\gamma$. This work scales on the full thickness H, with the Savva–Bush Ohnesorge number Oh and viscous time $\tau_{\mathrm{vis}} = \mu H/2\gamma$. The maps are

$$Oh_p = 2\,Oh, \qquad T = t^*/4, \qquad \mathcal{L} = \frac{\ell_0}{4h_0 Oh_p} = \frac{M}{4\,Oh}, \tag{1}$$

where $t^* = t/\tau_{\mathrm{vis}}$ and M is the total sheet mass per unit width, which is the extent of the computational domain in the mass coordinate. Their retraction speed $\mathrm{d}X_c/\mathrm{d}T$ equals $u^* =$

	TC1D-FEM	ABDS-CN	TC1D-MASS	TC1D-ALE
Model solved	full momentum balance, inertia retained	reduced form, inertia eliminated	reduced form, inertia eliminated	full momentum balance, inertia retained
Equations	$h_T = -h^2 u_m$, $u_T = 4\mu\,\partial_m(h^2 u_m)$	front-fixed heat equation of their Appendix D	$H_T = H^2 H_{mm}$	Eulerian form on $[x_c(t), \ell_0]$ with ALE correction
Unknown fields	two (h, u)	two (f, u)	one (H)	two (h, u) plus mesh position
Coordinate	Lagrangian mass	physical, algebraic stretching map	Lagrangian mass	physical, moving mesh
Discretisation	finite element, weak form; h cell-constant, u piecewise linear	finite difference, staggered grid	finite difference, nodal	finite element; h, u and mesh piecewise linear
Time integration	BDF2	Crank–Nicolson with Picard iteration	BDF2, Newton	BDF2
Linear algebra	symbolic Jacobian, sparse direct	tridiagonal (Thomas)	tridiagonal (Thomas)	symbolic Jacobian, sparse direct
Edge condition	natural: $h^2 u_m = -1/2\mu$ enters as a weak boundary term	$\surd$ tip shape imposed on the last two nodes	essential: $H(0,T) = 1 + T$	natural closure plus kinematic $\dot{x}_c = u(x_c)$ by Lagrange multiplier
Edge speed obtained as	nodal value of a solved field	one-sided gradient at the front	gradient of a Dirichlet solution	nodal value at the moving edge
Far field	$u = 0$, h free	clamped, Neumann thickness	$H = 1$	$u = 0$, h free

Table 1: The four solvers. Any two share at most one of the major choices. TC1D-FEM and TC1D-MASS share the mass coordinate but differ in the model integrated, in the number of unknowns, and in whether the edge condition is natural or essential; ABDS-CN works in a physical coordinate and so covers the case those two do not. TC1D-ALE shares the model and the framework with TC1D-FEM but places the sheet in the physical coordinate on a moving mesh, which is the discretisation the two-dimensional solver uses; it therefore tests that machinery on a problem with a known answer.

U/V_{TC} with no intervening factor. These relations were verified symbolically, and independently confirmed by two physical identities: their tip condition $HV_X = -1$ is the natural boundary condition of the present formulation, and their $H(X_c, T) = 1 + T$ is the tip-thickness relation measured here as $h_{\mathrm{tip}} = 1 + t^*/4$.

Equation (1) also shows that the present formulation represents a sheet of finite length: $\mathcal{L}$ is set by the mass domain M, and the far boundary condition (vanishing velocity, free thickness) is the condition imposed in the reference work. The mapping was checked before use: at $M = 20$, $Oh = 1$, the deceleration is observed at $T_c = 4.97$ against $\mathcal{L} = 5$.

2 Early time: $T \ll 1$

At early times the reference work predicts a diffusive growth of the retraction speed, $\mathrm{d}X_c/\mathrm{d}T \sim 2\sqrt{T/\pi}$ (their eq. 5.10). Figure 1 compares both solvers and that prediction at $\mathcal{L} = 0.1$, 1 and 20.

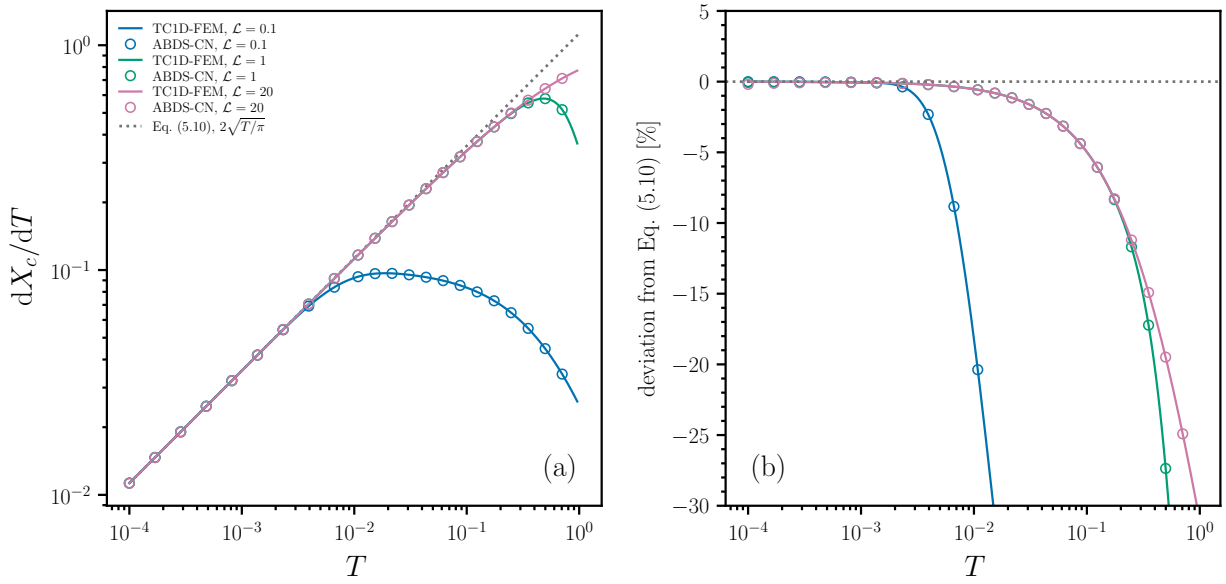


Figure 1: Retraction speed at early time. Solid: TC1D-FEM. Open symbols: ABDS-CN. Dotted: the diffusive prediction $2\sqrt{T/\pi}$.

The prediction is reproduced to better than 0.06% for $T \leq 10^{-3}$ at every $\mathcal{L}$. Its range of validity depends strongly on $\mathcal{L}$: taking the time at which the relative deviation first exceeds 1%, 5% and 10%,

	1%	5%	10%
$\mathcal{L} = 0.1$	$T = 0.0031$	0.0053	0.0075
$\mathcal{L} = 1$	$T = 0.019$	0.102	0.213
$\mathcal{L} = 20$	$T = 0.019$	0.102	0.221

The two solvers agree over this window to a maximum absolute difference of 2.4×10^{-5} at $\mathcal{L} = 0.1$, 1.3×10^{-4} at $\mathcal{L} = 1$ and 2.1×10^{-5} at $\mathcal{L} = 20$; the largest of these, 1.3×10^{-4}, is the figure quoted in the abstract.

3 Late time at finite length: the deceleration

When the free edge approaches the far end of the sheet, $\mathcal{L} - X_c \ll 1$, the reference work predicts $\mathrm{d}X_c/\mathrm{d}T \sim \mathcal{L}(1 + T)^{-2}$ (their eq. 6.10). This regime is the principal focus of that work.

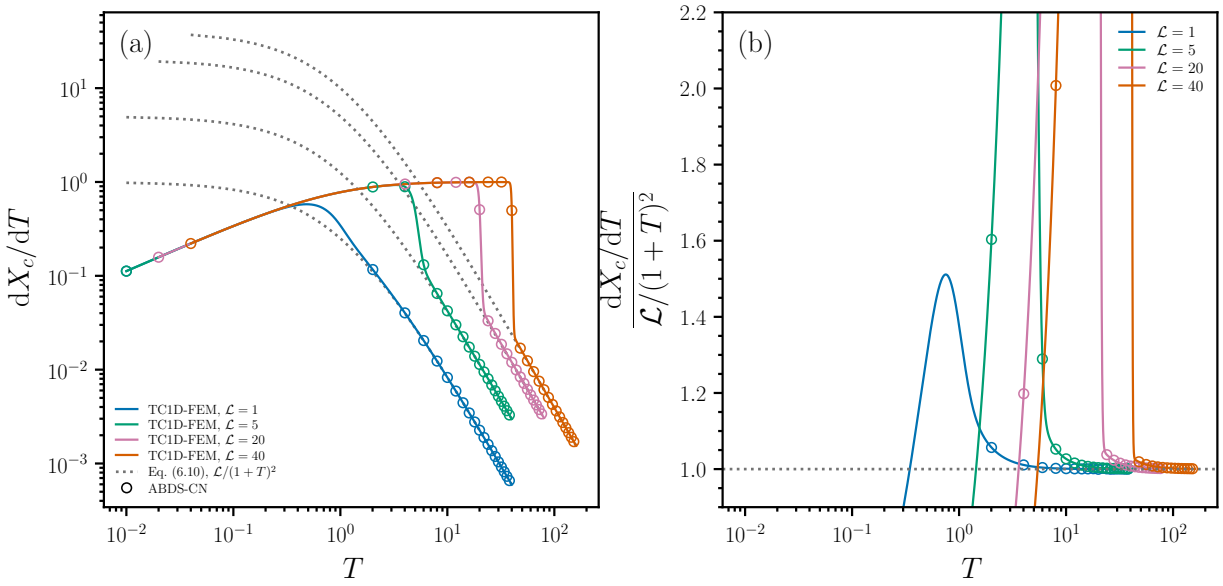


Figure 2: Deceleration at finite $\mathcal{L}$. Solid: TC1D-FEM. Open symbols: ABDS-CN. Dotted: $\mathcal{L}(1+T)^{-2}$.

The prediction is reproduced without any adjustable constant. At the end of each run the ratio of the computed speed to eq. (6.10) lies between 1.00002 and 1.0010, and the remaining sheet length $\mathcal{L} - X_c$ agrees with $\mathcal{L}/(1+T)$ to 1.3×10^{-4}. Continuing the $\mathcal{L} = 5$ case to $T = 400$ shows the relative departure decaying as $O(T^{-2})$, which is the order of the neglected terms.

4 Late time at large length

The regime of principal interest here is $\mathcal{L} \gg T \gg 1$, in which the edge has reached the Taylor–Culick speed but has not yet sensed the far end. The reference work gives, for an infinite sheet,

$$\frac{\mathrm{d}X_c}{\mathrm{d}T} \sim 1 - \frac{1}{(1+T)^2}. \tag{2}$$

Two points are worth stating before the comparison. The deficit is $(1+T)^{-2}$, not a pure power T^{-2}; fitting a power law in T and comparing exponents is therefore not a test of eq. (2). And the coefficient is described in that work as a numerical fit, with the corresponding expansion coefficient stated to remain undetermined by the asymptotic analysis and to depend on the history of the retraction.

4.1 The measured deficit

Figure 3 shows the speed and the deficit $1 - \mathrm{d}X_c/\mathrm{d}T$. Form the ratio

$$R(T) = (1+T)^2\Big(1 - \frac{\mathrm{d}X_c}{\mathrm{d}T}\Big), \tag{3}$$

which eq. (2) requires to approach unity. Three solvers, sharing no code, give instead

solver	model solved	$R(50)$
TC1D-FEM (finite element, mass coordinate)	momentum form	1.4986
ABDS-CN (finite difference, Richardson $\Delta X \to 0$)	reduced form	1.4985
TC1D-MASS (nonlinear diffusion, fixed domain)	$H_T = H^2 H_{mm}$	1.4986

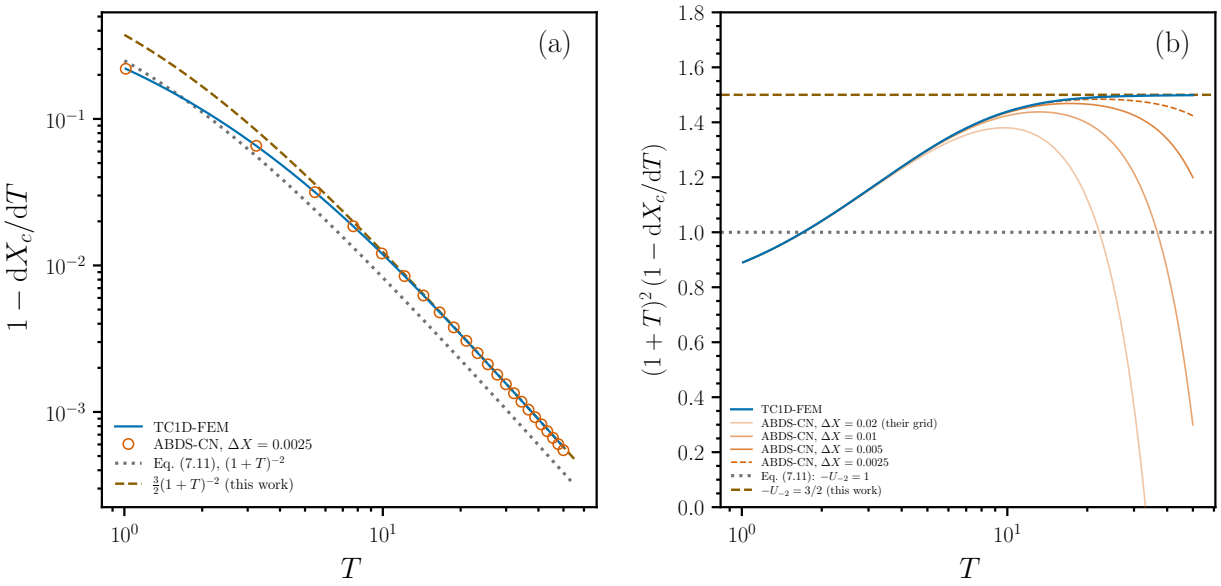


Figure 3: Late-time speed and deficit at large $\mathcal{L}$. Solid: TC1D-FEM. Open symbols: ABDS-CN. Dotted: eq. (2).

The third solver was written specifically to share nothing with the other two. In the mass coordinate the exact invariant reduces the problem to a nonlinear diffusion equation on a fixed domain, in which the free boundary disappears and the tip stress condition becomes an ordinary Dirichlet condition $H(0,T) = 1+T$. It therefore has a different coordinate, a different equation, a different boundary-condition type and no Stefan flux, and it agrees with the other two to six digits.

Continued to $T = 500$ at four timestep levels, with an observed convergence order of 1.993–1.998 throughout, the Richardson-extrapolated ratio is

T	50	100	200	300	400	500
R	1.498623	1.499660	1.499919	1.499993	1.500132	1.500319

R rises monotonically towards 3/2 and does not turn down. Up to $T \approx 300$ the residual $3/2 - R$ is positive and decays as $(1+T)^{-2}$; beyond that the tabulated values overshoot by a few parts in 10^4, which is the size of the timestep extrapolation there rather than a feature of the limit (the finest computed value at $T = 500$ reads 1.624 before extrapolation). We therefore quote the limit from the fit below rather than from the largest T reached. Writing $s = (1+T)^{-1}$ and $R = a_2 + a_3 s + a_4 s^2 + \ldots$, so that a_2 is the $s \to 0$ intercept, polynomial fits of degree one to three with the window start slid outward give an intercept stable at 1.50000, varying by less than 2×10^{-5} across window starts $T_{\min} = 50$ to 125. There is no drift toward unity. We take

$$\lim_{T\to\infty} (1+T)^2\Big(1 - \frac{\mathrm{d}X_c}{\mathrm{d}T}\Big) = 1.5000 \pm 0.0005, \tag{4}$$

i.e. 3/2 to within the quoted uncertainty, with unity excluded. We report this as a measurement rather than as an exact value; §4.4 shows it is a property of the initial condition. The same fits give $a_3 = 0.000 \pm 0.001$: there is no $(1+T)^{-3}$ term, so $(1+T)$ is the exact grouping, $U_{-3} = -2U_{-2} = 3$, and the next correction enters at $O\big((1+T)^{-4}\big)$.

The result is insensitive to every numerical parameter varied independently: mesh spacing (a factor of eight, changing $R(50)$ by 10^{-6}), domain truncation ($M = 70$, 90, 150, identical to every printed digit), timestep (two different step-control strategies, extrapolated limits differing by 3×10^{-5}), and, for ABDS-CN, the truncation length and ΔX.

Two further measurements of the same constant avoid the tip gradient entirely. From the displacement, $(X_c - T + 1)(1 + T)$ measures 1.4995 at $T = 50$; from the shape of the first correction to the thickness profile, away from the tip, 1.51 at $T = 50$.

4.2 The recommended form, and one that should be avoided

We therefore report

$$\frac{\mathrm{d}X_c}{\mathrm{d}T} = 1 - \frac{3}{2}\frac{1}{(1+T)^2} + O\big((1+T)^{-4}\big), \qquad X_c = T - 1 + \frac{3}{2}\frac{1}{1+T} + O(T^{-2}). \tag{5}$$

The single term is accurate to 0.09% over $50 \le T \le 200$ and 0.02% over $100 \le T \le 200$. Adding a $(1+T)^{-3}$ term to enforce the integral constraint makes the fit *worse* by more than an order of magnitude at large T, because it inserts a correction at an order where the data gives none; we recommend against it.

The displacement relation in eq. (5) deserves separate emphasis, because its constant is derivable and independent of the deficit coefficient. Expanding in the exact edge thickness $\theta = 1 + T$ gives $X_c = \theta - 2 + a_2/\theta$, whose $O(1)$ term is -1 for *any* a_2. Measurements give $T - X_c = 0.99255$ at $T = 200$ against a predicted 0.99254.

4.3 On the published coefficient

The integral constraint $\int_0^\infty (1 - \mathrm{d}X_c/\mathrm{d}T)\,\mathrm{d}T = 1$ is sometimes invoked to fix the coefficient. It cannot: the $O(1)$ term of X_c is -1 regardless of a_2, so the constraint is satisfied identically and carries no information about it. We verify this directly, recovering 1.00002 when the constraint is evaluated with the measured $a_2 = 3/2$.

The reference work's coefficient appears to follow from reading its uniform profile approximation (7.13) one order beyond its validity. That profile reproduces the leading two terms of the thickness expansion but not the third; substituting it into the integral relation (7.16) returns eq. (2) exactly. The same profile substituted into the alternative exact relation (3.11) returns a vanishing deficit. Two routes from one approximation disagreeing at precisely the order in question indicates that the order is beyond the approximation's reach.

4.4 Dependence on the initial condition

The coefficient is a property of the initial condition and not a universal constant of the model. Replacing the canonical rupture condition $H \equiv 1$, $V \equiv 0$ with $H(m, 0) = 1 + \delta e^{-m}$, which is admissible because the edge condition holds for any initial state, gives

δ	0 (canonical)	0.5	2
a_2	1.4997	1.8229	0.9366

The value 3/2 is therefore the canonical-rupture value. This supports the reference work's statement that the coefficient depends on the history of the retraction. The grouping $\theta = 1 + T$ and the result $a_3 = 0$ are unchanged across all three initial conditions.

4.5 Profiles and the tip thickness

Figure 4 compares thickness and velocity profiles against the reference work's eqs. (7.13) and (7.14). The tip-thickness relation $H(X_c, T) = 1 + T$ is reproduced to a maximum relative deviation of 3.6×10^{-5} over $1 \le T \le 50$. The profile predictions agree to 0.043% and 1.55% respectively at $T = 50$, improving monotonically with T.

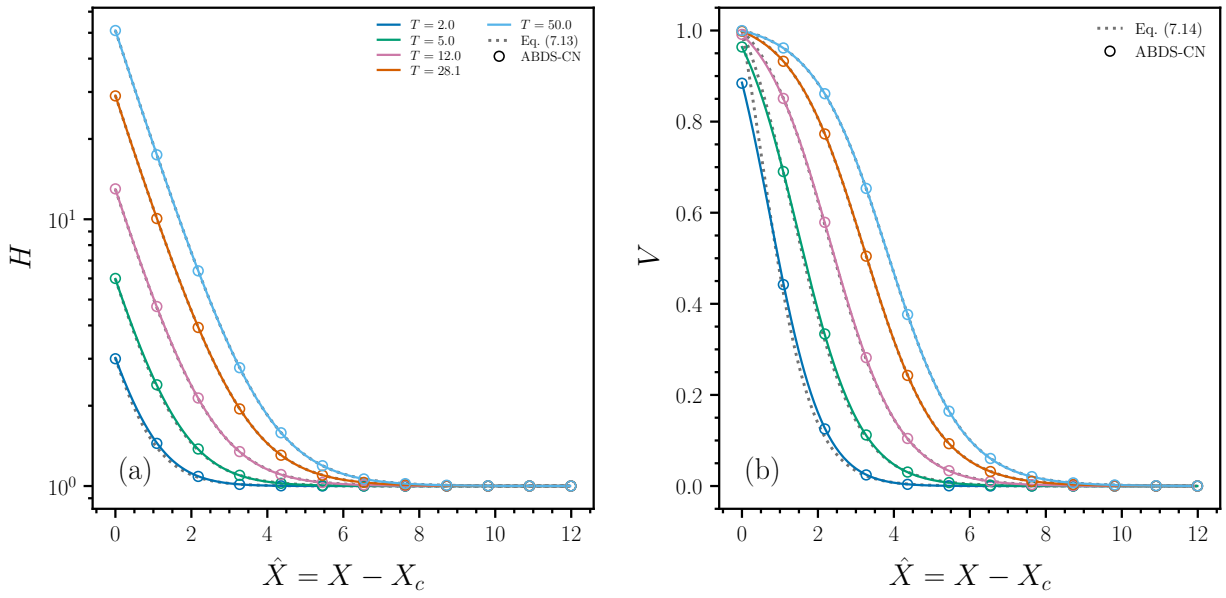


Figure 4: Thickness and velocity profiles at large $\mathcal{L}$. Solid: TC1D-FEM. Open symbols: ABDS-CN. Dotted: eqs. (7.13) and (7.14).

5 A moving-mesh solver

TC1D-ALE solves the same model as TC1D-FEM, with inertia retained, but in the physical coordinate on a mesh that moves. It exists to test the arbitrary Lagrangian–Eulerian machinery on a problem whose answer is known, because that machinery is what the two-dimensional solver relies on.

The mesh velocity w is neither the material velocity nor zero, so every time derivative taken at a fixed mesh point carries the convective correction $\mathrm{D}/\mathrm{D}t = \partial_t|_{\mathrm{mesh}} + (u - w)\partial_x$. At the free edge $w = u$, so the mesh tracks the interface; in the interior w is the affine stretch $w = u(x_c)(\ell_0 - x)/(\ell_0 - x_c)$, obtained as the solution of a mesh-motion equation rather than by renoding between steps. The edge carries the same natural closure as TC1D-FEM together with the kinematic condition $\dot{x}_c = u(x_c)$, imposed through a Lagrange multiplier because the boundary position depends on an unknown.

It reproduces all three reference quantities: $u^*(30)$ extrapolates to 0.9807279 in time and 0.9807278 in space, agreeing with each other to 1.5×10^{-7} and with TC1D-FEM's limit to 2.9×10^{-7}; the late-time ratio extrapolates to 1.49860, 1.49964 and 1.50000 at $T = 50$, 100 and 200; and the displacement relation holds with a residual of 10^{-7} relative. Observed convergence is second order in both time ($1.92 \to 1.99$) and space (2.000 at every refinement).

Two checks are worth recording. The mesh sits on the affine stretch to 6×10^{-15} of the domain length, so the mesh-motion solution is verified rather than assumed; and a control run with the same moving mesh but the ALE correction omitted drifts the total mass by a factor of sixteen. The correction is not a refinement but the difference between this result and none.

The physical coordinate loses resolution where the mass coordinate gains it: the rim occupies a fixed mass but a physical extent narrowing as $(1 + T)^{-1}$, while the affine stretch compresses the mesh only in proportion to the remaining sheet length. Late-time accuracy is therefore more expensive here than in TC1D-FEM. This is a property of the coordinate rather than of the implementation, and it is the reason the reference work adopts an algebraic stretching map.

6 The crossover time

For large $\mathcal{L}$ the edge decelerates from the Taylor–Culick speed onto the finite-sheet law of §3. The reference work reports the crossover time T_c and its duration ΔT_c as functions of $\mathcal{L}$, with $T_c = \mathcal{L}$ for large $\mathcal{L}$.

Neither quantity is defined in closed form in that work, so an operational definition is adopted here and applied identically to both solvers; it is stated with the figure. Figure 5 compares both solvers against values digitised from the published figure.

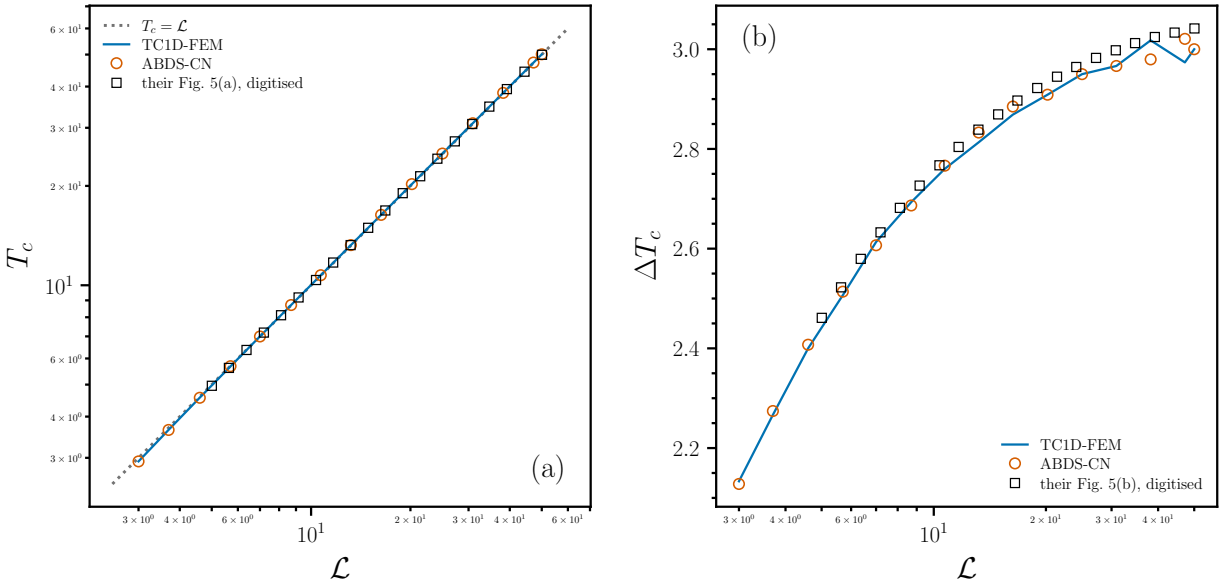


Figure 5: Crossover time T_c and duration ΔT_c against $\mathcal{L}$. Solid: TC1D-FEM. Open circles: ABDS-CN. Open squares: values digitised from the published figure.

We find $T_c/\mathcal{L} = 1$ to within 0.25% for $\mathcal{L} \geq 7$. Against the digitised values, T_c agrees to 0.142% RMS, within the estimated digitisation uncertainty of 0.2%, and ΔT_c to 0.024 absolute (1.5%).

One point of definition should be recorded. The text of the reference work describes ΔT_c as a full width at half maximum, while the released code computes twice that width. The published values are consistent with the code. Both conventions are reported here.

7 The range of validity assumed throughout

The four regimes examined above — the early response of §2, the terminal approach $\mathcal{L} - X_c \ll 1$ at arbitrary $\mathcal{L}$ of §3, the semi-infinite limit $\mathcal{L} \gg T \gg 1$ of §4, and the crossover at finite large $\mathcal{L}$ with $1 < T < \mathcal{L}$ of §6 — all share one assumption that is worth stating explicitly, because it is invisible in the equations actually solved.

The long-wave reduction carries no Ohnesorge number. That is what makes the curves of the preceding sections universal, and it is the reason a single solution serves every viscosity. The number has not disappeared from the physics, however: it has been absorbed into the time unit. The reduction remains faithful to the underlying flow only while the thin-film description does, and the reference work bounds that by $T = O(Oh_p)$, beyond which it identifies a two-dimensional inertial regime governed by the full Navier–Stokes equations. For the analysis of a sheet of length $\mathcal{L}$ to be uniformly valid over its whole history, one therefore needs

$$\mathcal{L} \ll Oh_p, \tag{6}$$

which in the units of (1) reads $M \ll 8\,Oh^2$. For a semi-infinite sheet $\mathcal{L} \to \infty$ and no finite viscosity satisfies (6); what survives is the weaker horizon $T \ll Oh_p$.

This bounds what the present validation can claim, and the boundary is worth being precise about. The agreement documented in §§2–6 is agreement between two discretisations of the same reduced model, and it is established to the precision reported there. It is not evidence that the reduced model describes a sheet at any particular Ohnesorge number, and no such claim is made. Establishing that requires a solver which never performs the reduction.

The question is therefore well posed and is left open here: for a semi-infinite sheet, how large must Oh_p be before a genuine two-dimensional flow spends a resolvable part of its history in the regime these sections describe? It cannot be answered from within the reduced model, which has already discarded the parameter the answer depends on. Preliminary direct simulations at $Oh_p \leq 10$ do not yet reach the asymptotic exponent of §4, which suggests the required viscosity is substantially larger than the range in which the long-wave model is usually invoked.

A Fidelity of the reference transcription

ABDS-CN is a transcription of released MATLAB into Python. To establish that the transcription is faithful rather than merely plausible, the original code was executed under GNU Octave and compared with the transcription at identical parameters. Over six configurations the largest absolute difference in the retraction speed is 5.0×10^{-10}, and the largest normalised difference across all twenty-four recorded quantities is 3.0×10^{-8}. The former is the resolution at which the original code writes its output, so the two agree to every digit that can be compared.

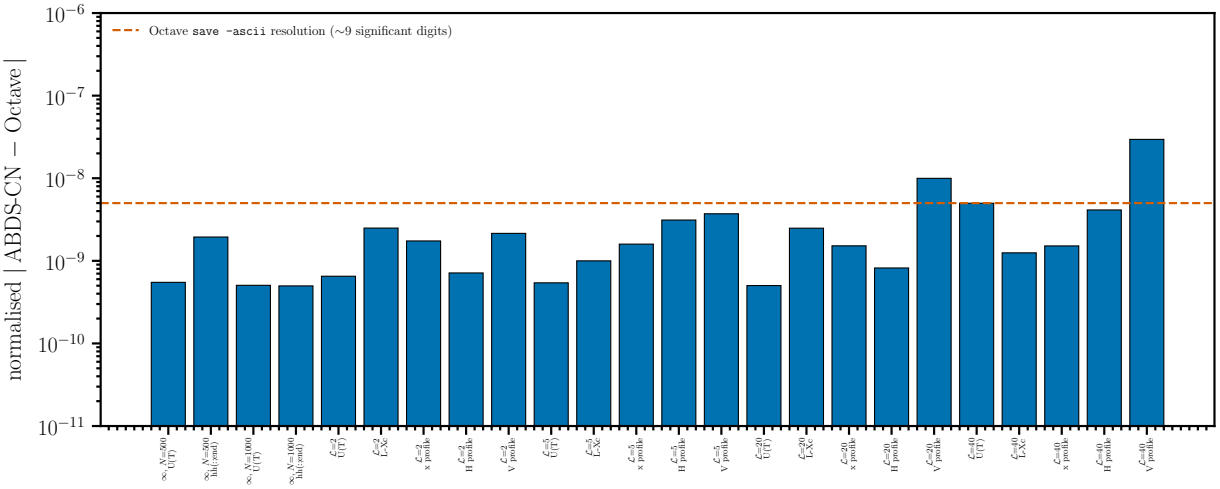


Figure 6: Original released code executed under GNU Octave, against the Python transcription ABDS-CN, at several $\mathcal{L}$.

B Reproduction

The campaign comprises 72 cases. Each is recorded in a manifest with its parameters and the exact command that regenerates it, alongside the derived series used for every figure and table in this document.